Модуль 2

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import tkinter as tk
from tkinter import ttk, messagebox
import psycopg2

DB_CONFIG = {
    "host": "localhost",
    "database": "practica",
    "user": "postgres",
    "password": "Nik12345",
    "port": "5432"
}

def get_connection():
    return psycopg2.connect(**DB_CONFIG)
def get_user_fio(login):
    conn = None
    try:
        conn = get_connection()
        with conn.cursor() as cur:
            cur.execute(
                "SELECT second_name, first_name, middle_name "
                "FROM users WHERE login = %s;",
                (login,)
            )
            row = cur.fetchone()
            if row:
                return f"{row[0] or ''} {row[1] or ''} {row[2] or ''}".strip()
    except Exception as e:
        messagebox.showerror("Ошибка БД", str(e))
    finally:
        if conn:
            conn.close()
    return None
def fetch_users_rows():
    conn = None
    try:
        conn = get_connection()
        with conn.cursor() as cur:
            cur.execute("""
                SELECT u.user_id, u.first_name, u.second_name, u.middle_name,
                       u.login, r.role_name
                FROM users u
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                LEFT JOIN roles r ON u.role_id = r.role_id;
            """)
            return cur.fetchall()
    except Exception as e:
        messagebox.showerror("Ошибка БД", str(e))
        return []
    finally:
        if conn:
            conn.close()

def clear_root():
    for w in root.winfo_children():
        w.destroy()

def show_login_screen():
    clear_root()
    root.title("Вход в систему")
    root.geometry("400x260")
    root.resizable(False, False)
    tk.Label(
        root,
        text="Магазин BURMOLDA",
        font=("Arial", 18, "bold")
    ).pack(pady=(25, 15))
    tk.Label(root, text="Логин пользователя", font=("Arial", 11)).pack()
    entry_login = tk.Entry(root, width=25, font=("Arial", 12))
    entry_login.pack(pady=6)
    entry_login.focus()
    def handle_login():
        login = entry_login.get().strip()
        if not login:
            messagebox.showwarning("Ошибка", "Введите логин!")
            return
        fio = get_user_fio(login)
        if fio:
            show_main_screen(fio)
        else:
            messagebox.showerror("Ошибка", "Пользователь не найден!")
    entry_login.bind('<Return>', lambda e: handle_login())
    tk.Button(
        root,
        text="Войти в систему",
        command=handle_login,
        font=("Arial", 11),
        relief="solid",
        borderwidth=2,
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        width=18,
        ipady=4
    ).pack(pady=18)

def show_main_screen(user_fio):
    clear_root()
    root.title("Главная страница")
    root.geometry("1000x650")
    root.resizable(True, True)
    tk.Label(
        root,
        text=f"Добро пожаловать, {user_fio}!",
        font=("Arial", 18, "bold")
    ).pack(pady=(20, 5))
    tk.Label(
        root,
        text="Магазин BURMOLDA — каталог товаров",
        font=("Arial", 13),
        fg="#555"
    ).pack(pady=(0, 10))
    nav = tk.Frame(root)
    nav.pack(pady=5)
    tk.Button(
        nav, text="Каталог", font=("Arial", 11), width=14,
        command=lambda: show_catalog_screen(user_fio)
    ).pack(side="left", padx=5)
    tk.Button(
        nav, text="Пользователи", font=("Arial", 11), width=14,
        command=show_users_screen
    ).pack(side="left", padx=5)
    tk.Button(
        nav, text="Выйти", font=("Arial", 11), width=14,
        bg="#f44336", fg="white",
        command=show_login_screen
    ).pack(side="left", padx=5)
    tk.Label(
        root, text="Выберите раздел выше.",
        font=("Arial", 12), fg="#777"
    ).pack(expand=True)

def show_catalog_screen(user_fio):
    clear_root()
    root.title("Каталог")
    root.geometry("1000x650")

    top = tk.Frame(root)
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    top.pack(fill="x", pady=(15, 5))
    tk.Button(
        top, text="← Назад", font=("Arial", 11),
        command=lambda: show_main_screen(user_fio)
    ).pack(side="left", padx=10)
    tk.Label(
        top, text="Магазин BURMOLDA — каталог",
        font=("Arial", 18, "bold")
    ).pack(side="left", expand=True)

    canvas = tk.Canvas(root, borderwidth=0, highlightthickness=0)
    scrollbar = ttk.Scrollbar(root, orient="vertical", command=canvas.yview)
    cards_frame = tk.Frame(canvas)
    cards_frame.bind(
        "<Configure>",
        lambda e: canvas.configure(scrollregion=canvas.bbox("all"))
    )
    canvas.create_window((0, 0), window=cards_frame, anchor="nw")
    canvas.configure(yscrollcommand=scrollbar.set)
    canvas.pack(side="left", fill="both", expand=True, padx=10, pady=10)
    scrollbar.pack(side="right", fill="y")
    products = [
        {"producer": "Россия", "name": "Кроссовки", "category": "Обувь",
         "qty": "10", "composition": "Кожа", "price": "4999 ₽"},
        {"producer": "Китай", "name": "Футболка", "category": "Одежда",
         "qty": "25", "composition": "Хлопок", "price": "1299 ₽"},
        {"producer": "Турция", "name": "Куртка", "category": "Верхняя одежда",
         "qty": "5", "composition": "Полиэстер", "price": "7999 ₽"},
    ]
    for p in products:
        create_card(cards_frame, p)
def create_card(parent, product):
    card = tk.Frame(
        parent,
        highlightbackground="black",
        highlightthickness=2,
        bd=0
    )
    card.pack(pady=10, padx=20, fill="x")
    photo = tk.Frame(
        card, width=100, height=100,
        highlightbackground="black", highlightthickness=2
    )
    photo.pack(side="left", padx=15, pady=15)
    photo.pack_propagate(False)
    tk.Label(
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        photo, text="ФОТО", font=("Arial", 14, "bold")
    ).place(relx=0.5, rely=0.5, anchor="center")
    info = tk.Frame(card)
    info.pack(side="left", fill="both", expand=True, padx=10, pady=10)
    top_row = tk.Frame(info)
    top_row.pack(anchor="w", pady=(0, 8))
    tk.Label(top_row, text=product["producer"],
             font=("Arial", 14, "bold")).pack(side="left")
    tk.Label(top_row, text=" | ",
             font=("Arial", 14, "bold")).pack(side="left")
    tk.Label(top_row, text=product["name"],
             font=("Arial", 14, "bold")).pack(side="left")
    for key in ["category", "qty", "composition"]:
        tk.Label(info, text=product[key], font=("Arial", 13),
                 anchor="w").pack(fill="x", pady=1)
    tk.Label(info, text=f"Цена: {product['price']}",
             font=("Arial", 14, "bold"),
             anchor="e").pack(fill="x", pady=(8, 0))

def show_users_screen():
    clear_root()
    root.title("Пользователи")
    root.geometry("900x450")
    tk.Button(
        root, text="← Назад", font=("Arial", 11),
        command=lambda: show_login_screen()
    ).pack(anchor="w", padx=10, pady=(10, 0))
    tk.Label(
        root, text="Список пользователей",
        font=("Arial", 14, "bold")
    ).pack(pady=5)
    columns = ("user_id", "first_name", "second_name",
               "middle_name", "login", "role_name")
    tree = ttk.Treeview(root, columns=columns, show="headings")
    headers = {
        "user_id": "ID",
        "first_name": "Имя",
        "second_name": "Фамилия",
        "middle_name": "Отчество",
        "login": "Логин",
        "role_name": "Роль"
    }
    widths = {"user_id": 60, "first_name": 120, "second_name": 120,
              "middle_name": 120, "login": 130, "role_name": 130}
    for col in columns:
        tree.heading(col, text=headers[col])
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        tree.column(col, width=widths[col],
                    anchor="center" if col == "user_id" else "w")
    tree.pack(fill="both", expand=True, padx=10, pady=5)
    def refresh():
        for item in tree.get_children():
            tree.delete(item)
        for row in fetch_users_rows():
            tree.insert("", "end", values=row)
    tk.Button(root, text="Обновить", command=refresh,
              font=("Arial", 12)).pack(pady=8)
    refresh()

if __name__ == "__main__":
    root = tk.Tk()
    show_login_screen()
    root.mainloop()
```